# ISSAC PAUL

CHANDANATHIL HOUSE,
THIRUVANKULAM P.O, ERNAKULAM,
KERALA, INDIA 682305
PHONE: +91 8547016405
EMAIL: issacpaul45@gmail.com

**GITHUB:** https://github.com/issacpaul45

## PROFILE SUMMARY

Backend-focused Software Engineer with proven experience building and maintaining production applications using Python, Django, PostgreSQL, Celery, RabbitMQ, Docker, and AWS. Experienced in REST API development, asynchronous processing, performance optimization, production debugging, root-cause analysis, authentication, and role-based access control.

## TECHNICAL SKILLS

| Category | Skills |
|---|---|
| **Programming Languages** | Python, JavaScript, TypeScript |
| **Backend Development** | Django, Flask, REST APIs, Socket Programming, JWT Authentication, Role-Based Access Control, Celery, RabbitMQ, Asynchronous Task Processing, API Integration, LLM API Integration, Geospatial Processing, Algorithmic Optimization |
| **Databases** | PostgreSQL, MySQL, SQLite3, MongoDB |
| **Tools and Platforms** | Git, GitHub, Postman, V S Code |
| **Architecture and Concepts** | Monolithic Architecture, Queue-Based Architecture, Event-Driven Architecture, Background Jobs, Notification Systems |
| **Cloud / DevOps** | AWS Basics, Docker, Linux Basics, Deployment Support, Production Debugging |

## WORK EXPERIENCE

### INDAPP

### Software Engineer Level 1 (October 2025 - Present)

#### Roles and Responsibilities

- Developed and maintained production backend services and RESTful APIs using **Python, Django, and PostgreSQL** for an application serving **thousands of users**, applying software design patterns and object-oriented principles.
- Designed and implemented secure REST APIs with **JWT authentication and role-based access control (RBAC)**, supporting reliable production services.
- Built asynchronous processing pipelines using **Celery and RabbitMQ** for notifications and background tasks, improving workload distribution, and reducing processing load on synchronous API requests.
- **Optimized API performance** by refactoring serializer processing that has become a bottleneck, reducing unnecessary API load, and implementing database-level improvements to address performance issues.
- **Investigated and resolved production issues through root-cause analysis**, identifying underlying application and database problems, and implementing preventive fixes to reduce recurrence.

- Containerized **Django, Celery, and PostgreSQL** services using **Docker and Docker Compose**, standardizing local development and deployment workflows.
- Integrated **AWS S3, SQS (FIFO), and IAM** into Django applications for secure cloud storage, queue-based processing, and controlled service access.
- Owned features across the **software development lifecycle**, from design and implementation through deployment and production debugging, while collaborating with QA and DevOps.
- Managed version control and code quality using **Git and GitHub**, including pull requests, branch management, and peer code reviews.
- Led knowledge transfer sessions for contract developers on legacy system maintenance.

**QUEST INNOVATIVE SOLUTIONS, KOCHI**

**Jr.Software Engineer (March 2024 – April 2025)**

## Roles and Responsibilities

- Spearheaded development of **full-stack** applications using Django, Angular, Python, and RESTful APIs.
- Mentored and trained multiple interns in web development fundamentals and best practices.
- Collaborated with cross-functional teams to deliver high-quality software solutions.

## PROJECTS

**Route & Fuel Optimization API | Backend Developer**
**Django, DRF, PostgreSQL, Docker, Geoapify, Shapely**

- Built an API that calculates optimal fuel stops along US routes based on fuel prices, vehicle range, fuel efficiency, and route distance.
- Integrated external geocoding/routing APIs with geospatial processing and route-based fuel-station discovery.
- Implemented caching, optimization logic, structured error handling, and automated tests.
- Dockerized the application with PostgreSQL and followed a Git/GitHub development workflow.

**College Management System | Backend Developer**
**Django, SQLite3, REST APIs, Razorpay**

- Architected and developed a college management platform using Django and SQLite3.
- Implemented authentication and role-based access control (RBAC) for students, faculty, and administrators.
- Designed RESTful APIs for academic data, course scheduling, and student records.
- Integrated Razorpay in demo mode for online fee payments as a proof of concept.

**LLM Chatbot**
**Python, OpenAI API**

- Built a conversational chatbot using the **OpenAI API** and Python.
- Implemented multi-turn conversations by maintaining conversation history.
- Added environment-based API key management and API error handling.
- Used the OpenAI Python SDK to integrate LLM responses into the application.

## ACADEMIC QUALIFICATION

- **Bachelor of Technology** (Computer Science Engineering) – 2023:
  TOC H Institute of Science and Technology, Kochi, APJ Abdul Kalam Technological University (KTU), Kerala, India